

MELINA BAGHER

Tehran-Iran

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🌐 Personal Website

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EDUCATION

🎓 MASTER OF SCIENCE IN BIOMEDICAL ENGINEERING

Sep. 2019 – Sep. 2022

Sharif University of Technology

Tehran, Iran

- **Thesis:** “Comparative Analysis of Haplotype Assembly Algorithms to Identify Optimal Methods”
- **Advisor:** Prof. Babak Hosseini Khalaj, Prof. Mehran Jahed
- **GPA:** 19.19/20 → 4/4 (Ranked)

🎓 BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING

Sep. 2014 – Sep. 2018

Islamic Azad University, Science and Research Branch

Tehran, Iran

- **Thesis:** “Extraction of Fractal Dimension of Malignant and Benign Skin Lesions in Dermoscopic Images”
- **Advisor:** Prof. S. Rashidi
- **GPA:** 19.10/20 → 4/4 (Ranked)

PUBLICATIONS

📖 JOURNAL PAPERS

○ Bagher M., Karimzadeh R., khalaj B.H., Jahed M; *A rapid heuristic algorithm to solve the single individual Haplotype assembly problem* ; AUT Journal of Electrical Engineering 55 (2), 191-206, 2023.

📖 CONFERENCE PAPERS

○ Bagher M., Karimzadeh R., khalaj B.H., Jahed M; *QuickHap: a Quick heuristic algorithm for the single individual Haplotype reconstruction problem* ; 29th National and 7th International Iranian conference on Biomedical Engineering 2022 (ICBME).

○ Karimzadeh R., Bagher M., Khodabakhsh A., Arabi H., Zaidi H; *Generative Adversarial-based Framework for Classification using Imbalance Data: Application to Pneumonia Detection in Chest Radiographs*; IEEE NSS and Medical Imaging Conference 2022.

SELECTED ACADEMIC PROJECTS

📖 Artificial Intelligence and Deep Learning Projects/Tasks

☆ CycleGAN for Image-to-Image Translation

Developed a CycleGAN model using PyTorch for unsupervised image-to-image translation between two distinct domains, training two generators and two discriminators. Optimized the model's performance to ensure high-quality image transformations.

📝 Pursued out of personal interest in generative models.

☆ Enhancing Adversarial Robustness in Convolutional Neural Networks

Developed a convolutional neural network using PyTorch to evaluate and improve robustness against adversarial attacks. Applied Projected Gradient Descent (PGD) and Fast Gradient Sign Method (FGSM) for generating adversarial examples and optimized the model for both robust and clean accuracy.

✍ *Part of the "Trustworthiness in DeepLearning" course, Rayan International AI Contest*

☆ Image Generative Models: VAE, GAN, and Stable Diffusion Pipeline Implementation and Analysis

Implemented and analyzed latent diffusion pipelines based on Stable Diffusion, including U-Net-based denoising, VAE-based decoding, and text conditioning mechanisms. Explored the role of latent representations and noise scheduling in diffusion-based text-to-image generation.

✍ *Part of a personal project exploring AI-driven creative tools.*

☆ Semi-Supervised GAN for Pneumonia Detection

Developed a semi-supervised GAN model using PyTorch to classify pneumonia in chest X-ray images. The model enhanced classification accuracy by generating synthetic features and integrating both GAN and classification loss, improving performance on imbalanced datasets.

✍ *Voluntarily pursued due to personal interest, resulting in a publication listed earlier in my resume.*

☆ Bone fracture classification using multi-level feature fusion

Developed a transfer learning-based bone fracture classification model by combining features extracted from multiple intermediate layers of InceptionV3 and integrating them into a custom-designed classification head, achieving improved performance over standard single-layer transfer learning models.

✍ *Self-initiated project motivated by interest in transfer learning*

☆ License Plate Recognition using CNN-based Sequence Modeling

Developed a CNN-based model for Iranian license plate recognition, performing fixed-length character-wise classification directly from input images using supervised learning.

✍ *Self-initiated project*

📁 Medical Image Processing Projects/Tasks

☆ Development of Registration Pipeline for Spinal Cord CT Images

Developed a pipeline using Coherent Point Drift (CPD) and Iterative Closest Point (ICP) algorithms in MATLAB for accurate registration of spinal cord CT images. Evaluated the results using Hausdorff Distance (HD), Dice Score (DS), and Average Surface Distance (ASD).

✍ *Medical Image Processing and Analysis Course Project*

☆ Preprocessing, Artifact Elimination, and Skin Lesion Segmentation of Dermoscopy Images

Utilized MATLAB to remove noise, hair, and air bubbles from dermoscopy images through various filtering and morphological techniques. Subsequently, skin lesions were segmented using multiple methods to enhance diagnostic accuracy.

✍ *In line with my BSc project*

📁 Computational Genomics and Bioinformatics Projects/Tasks

☆ A convolutional neural network for Single Individual Haplotype Reconstruction

Designed a single-layer convolutional neural network using PyTorch to address the haplotype assembly problem. Developed a novel cost function to enhance accuracy and improve the overall performance of the model.

✍ *In line with my final project*

☆ NGS (Next Generation Sequencing) data simulation and variant calling

Utilized tools such as PBSim and NanoSim to simulate sequencing data in Fastq format. Implemented a variant calling pipeline using BWA-MEM, GATK, Samtools, and BCFtools to prepare for subsequent analyses.

 *In line with my final project*

☰ Other Projects, Tasks and Simulations

☆ Simulation and Comparison of SFP and MFP Transducers

Designed and simulated Single-Focal Point (SFP) and Multi-Focal Point (MFP) transducers in MATLAB, comparing Lateral Resolution and Depth of Field (DOP) across frequencies to identify the optimal structure for ultrasound applications.

 *Medical Ultrasound Course Project*

HONORS & AWARDS

- Ranked 1st among approximately 500 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering (Bioelectric), 2023
- Ranked 2nd among approximately 900 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering, 2026
- Ranked 39th among approximately 40,000 participants in the nationwide M.Sc. entrance exam for Electrical Engineering
- Member of National Elite Foundation
- Distinguished student in B.Sc degree

TEACHING EXPERIENCE

- **TEACHING ASSISTANT** | Medical Images Analysis and Processing | Supervisor: Prof. Emad Fatemizadeh
Department of Electrical Engineering, Sharif University of Technology *Fall 2021*
- **TEACHING ASSISTANT** | System Biology | Supervisor: Prof. Babak Hossein Khalaj
Department of Electrical Engineering, Sharif University of Technology *spring 2020*
- **TEACHING ASSISTANT** | Electronics 2 | Supervisor: Prof. Keyvan Maghouli
Department of Biomedical Engineering, Islamic Azad University, Science and Research Branch *Fall 2019*
- **TUTOR** | Engineering Mathematics, Differential Equations, Signals and Systems,
Mahan Institute of Higher Education *Summer 2019*

SELECTED COURSES

☰ Academic Relevant Courses:

☆Medical Image Analysis and processing: Grade 19.7/20 (4/4)

Sharif University of Technology, 2021

Instructor: Emad. Fatemizadeh

☆ Biomedical Signal Processing: : Grade 18.9/20 (4/4)

Sharif University of Technology, 2021

Instructor: Mohammad B. Shamsollahi

📖 Online Extracurricular Courses

☆ Supervised Machine Learning; Regression and Classification

DeepLearning.AI, Stanford University via Coursera, Jan. 2024

Instructor:Andrew Ng

☆ Advanced Learning Algorithms

DeepLearning.AI, Stanford University via Coursera, Jan. 2024

Instructor:Andrew Ng

☆ Introduction to Machine Learning and DeepLearning

Rayan.AI, Sharif University of Technology, Sep. 2024

Instructor:Mohammad Hossein Rohban, Rayan.AI Group

☆ Trustworthiness in DeepLearning

Rayan.AI, Sharif University of Technology, Sep. 2024

Instructor:Mohammad Hossein Rohban, Rayan.AI Group

TECHNICAL SKILLS

Programming Languages:	Python, R, C/C++, Bash
Python Selected Libraries:	Pytorch, TensorFlow, Scikit-learn, Matplotlib, Numpy, Scipy
Version Control:	Git, GitHub
Software Simulators:	Matlab, Simulink
Hardware Simulator:	Pspice, Proteus
Typesetting:	T _E X, Prezi, Microsoft Office(Word, Powerpoint, Excel)
Operating Systems:	Windows,Ubuntu
NGS Tools and Simulators:	GATK Toolkit, BWA, Samtools, BCFtools, PBSIM, NanoSim, ART

RESEARCH INTERESTS

- Medical Image Processing, Computer Vision
- Medical Signal Processing
- Machine learning, Deep Learning
- Computational Genomics, NGS data analysis

LANGUAGES

English TOEFL-iBT: 98 (R:28, L:21, S:23, W:26)

French A1

Persian/Farsi Native

HOBBIES

SPORTS: Volleyball, Badminton, Ping-pong

ART: Piano Player (Advanced), Guitar Player (Intermediate)

WORKING EXPERIENCE

Medical Equipments unit of Parsian Hospital

Summer 2018

ENGINEERING INTERNSHIP

- Becoming familiar with Repair and maintenance of health monitoring equipments ...

Educational counsellor

2019 - 2021

Mahan Institute of Higher Education

- Consulting to Master and Ph.D Konkoor Candidates.

Private Piano Tutor

2016-Present

Teaching piano to students of various age groups and skill levels

REFERENCES

Asst. Prof. Saeed. Rashidi, Department of Biomedical Engineering, Islamic Azad University, Science and Research Branch

Email: rashidi@srbiau.ac.ir

Prof. Babak Hosseini Khalaj, Department of Electrical Engineering, Sharif University of Technology

Email: khalaj@sharif.edu

Asst. Prof. Mehran Jahed, Department of Electrical Engineering, Sharif University of Technology

Email: Jahed@sharif.edu